

Real-Time Decoder and Player for MPEG V-DMC Dynamic Meshes

Alekssei Martemianov

Lauri Ilola

Emre B. Aksu

Lukasz Kondrad

Email: {firstname.lastname}@nokia.com
Nokia Technologies

Abstract— This paper presents a real-time decoder and player for 3D dynamic meshes, compressed using the MPEG V-DMC standard (Video-based Dynamic Mesh Coding). MPEG V-DMC efficiently compresses dynamic 3D meshes, enabling smooth playback of high-resolution meshes with minimal latency. We demonstrate that the MPEG V-DMC standard is highly effective for representing dynamic meshes, offering significant data compression while maintaining high visual quality. Our implementation further shows that MPEG V-DMC can be successfully deployed on simple, resource-constrained devices, making it ideal for a wide range of applications, including volumetric media streaming, augmented reality (AR), and virtual reality (VR), as well as more traditional computer graphics workflows. By leveraging modern web technologies, we provide a platform-independent solution that broadens the accessibility of high-quality XR content.

Keywords— *dynamic mesh coding, mesh compression, volumetric streaming, real-time rendering, WebXR, AR/VR applications.*

I. INTRODUCTION

Dynamic mesh sequences are now widely used in communication, gaming, extended reality (XR), augmented reality (AR), and general computer graphics applications. For such scenarios, achieving low latency is crucial to maintain user comfort and immersion, which underscores the importance of advanced compression and real-time decoding solutions. These sequences present detailed 3D objects and dynamic environments, allowing for immersive and interactive user experiences.

With the increasing complexity of immersive applications, such as AR, VR, and XR, and more conventional computer graphics scenarios, earlier approaches struggled to balance quality, performance, and scalability. MPEG V-DMC standard[2] was designed to compress high-resolution 3D content with dynamic changes, making it suitable for use on a wide range of devices, including mobile platforms.

II. MPEG V-DMC OVERVIEW

The MPEG Video-based Dynamic Mesh Coding (V-DMC) standard structures volumetric data into metadata streams that facilitate 3D reconstruction, alongside video sub-streams that capture geometry and attributes such as texture, colors, and normals. Unlike dynamic point clouds, dynamic meshes requires additional connectivity information. To address this, MPEG V-DMC employs a base mesh concept—a simplified and decimated representation of the original mesh [3,4].

During reconstruction, the decoded sub-streams (meta data, base mesh, displacement, and attributes) are utilized to reassemble the 3D mesh frames. This process involves subdividing the base mesh to approximate the original mesh, with displacement values indicating the differences between

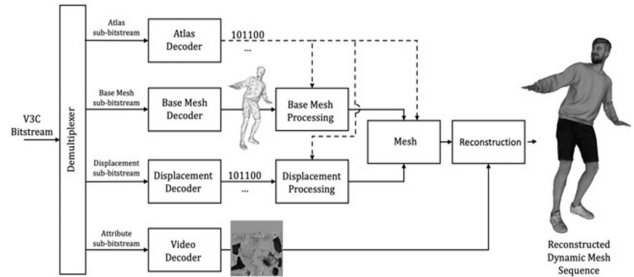


Figure 1: MPEG V-DMC decoder scheme

the original and subdivided meshes at the subdivision points. Subsequently, a texture (and other attributes) map are applied to reconstruct the complete 3D mesh frame. Motion information represents difference between base-meshes.

III. MPEG V-DMC PLAYER IMPLEMENTATION

A web-based application was developed for real-time playback of MPEG V-DMC encoded dynamic meshes. The selection of the web platform was driven by several key advantages:

- ✓ **Platform-agnostic:** “Write once run everywhere” is the basic principle of web-platform.
- ✓ **Hardware Utilization:** Leveraging GPU shaders and WebCodecs allows optimal use of device hardware.
- ✓ **Integration Capabilities:** Easily integrates with AR frameworks, XR, web services, and multimedia content for an enhanced user experience.

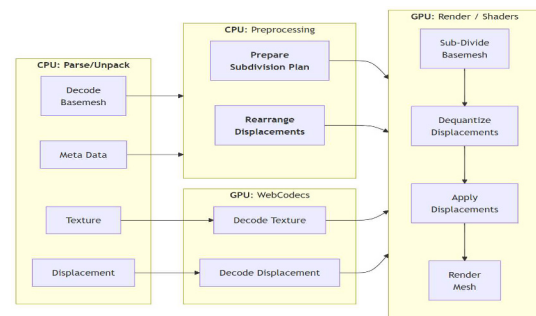


Figure 2: MPEG V-DMC Player high-level processing chain.

Figure 2 illustrates the real-time processing pipeline for MPEG V-DMC dynamic meshes, combining CPU-based parsing and preprocessing[1] with GPU-accelerated decoding and rendering. The CPU handles basemesh decoding, metadata extraction, and preparation of subdivision and displacement data. Once prepared, the GPU performs hardware-accelerated decoding of textures and displacements using WebCodecs (HEVC).

The final stage involves subdividing the basemesh triangles, dequantizing and applying displacements, and rendering the complete 3D mesh by using WebGL2 shaders.

IV. RESULTS AND EVALUATION

The implementation of MPEG V-DMC decoder and player demonstrates that it is possible to decode MPEG V-DMC encoded meshes and render them in real-time, even in a web environment.



Figure 3: Demo playback in AR [5]

The implementation is optimized to minimize latency and maximize efficiency of GPU loading, leveraging parallel processing capabilities of modern CPU and GPUs.

We analyzed the performance of the V-DMC decoder executed on a target device (Table 1) with a simple profile sequence and highlighted potential bottlenecks (Table 2) Despite these challenges, the decoder demonstrates reliable real-time performance, effectively balancing resource constraints and high-quality rendering for immersive applications. WebXR currently does not support WebGPU, limiting us to only WebGL2 shaders, which, on the other hand, improves portability and accessibility for all devices.

OS	Android™ 14
CPU	Qualcomm SM8550-AB Snapdragon 8 Gen 2 (4 nm), Octa-core
GPU	Adreno 740 (700 MHz, 2089.0 GFLOPs)
RAM	8GB LPDDR5X
APIs	HTML5/JavaScript/WebAssembly, WebGL2/WebXR, WebCodecs (HW HEVC)

Table 1: Demo device specification

CPU (JS/WASM) Parsing/Decoding	Base-mesh can be decoded using a single CPU core. For modern devices it gives 4-5 base-meshes in realtime.
GPU (WebGL2) Rendering	Up to 40 million triangles per second can be rendered, including: - Fetching displacements, - Inverse quantization, - Subdivision, - Shading. This is enough to decode up to 8 V-DMC streams @ 30 FPS (~150,000 triangles per frame).
HW Decoding (HEVC)	The hardware HEVC decoder with resolutions of 2048x2048 for textures and 256x512 for displacements is able to process up to 5 streams in real-time.

Table 2: Performance details

V. CONCLUSIONS AND FUTURE WORK

We demonstrated a real-time decode and playback system for dynamic meshes encoded in MPEG V-DMC standard, which offers efficient compression and high-quality rendering, suitable for applications in 3D streaming, XR, and AR/VR environment. This prototype shows that the MPEG V-DMC standard is implementable for a web platform and can run on existing mobile and desktop hardware in browser.

Future work will focus on deeper optimization to expanding support for more features and higher resolution meshes. With official availability of WebGPU for WebXR, the efficiency and flexibility of the GPU optimizations can be improved to support more features of the MPEG V-DMC standard.

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